

Can World Models Optimize Customer Journeys and lifetime value?

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Abstract

Customer-journey optimization and customer lifetime value (CLV) management are sequential decision problems: actions taken today (emails, recommendations, discounts) change future engagement, fatigue, and churn risk. World models—learned simulators of environment dynamics—enable multi-step planning under delayed effects, but can fail when trained on confounded business-as-usual logs. We propose a diagnostic simulation framework that separates training (learning a world model) from grading (evaluating policies in a ground-truth simulator). In a controlled environment with delayed costs, we compare world-model planning under randomized vs confounded logging regimes and quantify a regime-dependent “reality gap” between simulated and true policy value.

1 Introduction

Optimizing CLV requires trading off short-term conversions against long-term retention. One-step response models (uplift, immediate conversion probability) ignore delayed effects such as fatigue and churn. World models aim to learn customer dynamics from logged data and support multi-step planning by imagining futures under candidate action sequences.

2 Diagnostic setup

We use a Level-0 fully observable environment with state $s_t = (E_t, F_t, P_t) \in [0, 1]^3$ capturing engagement, fatigue, and purchase propensity. Actions are `{none, email, reco, discount}`. Reward is revenue-focused:

$$r_t = 100 \cdot \mathbb{I}[\text{purchase}] - c(a_t) - 300 \cdot \mathbb{I}[\text{churn}],$$

with proportional discount cost $c(a_t)$ for `discount`. We train a world model to predict $(s_{t+1}, p_{\text{buy}}, p_{\text{churn}})$ from (s_t, a_t) .

We compare two logging regimes:

- **Randomized logs:** actions sampled independently of state (broad coverage).
- **Confounded logs:** business-as-usual heuristics assign actions based on state (selection bias).

Planning uses MPC: sample candidate sequences of length H , roll out in $\hat{P}$, execute the first action in the ground-truth environment, and repeat.

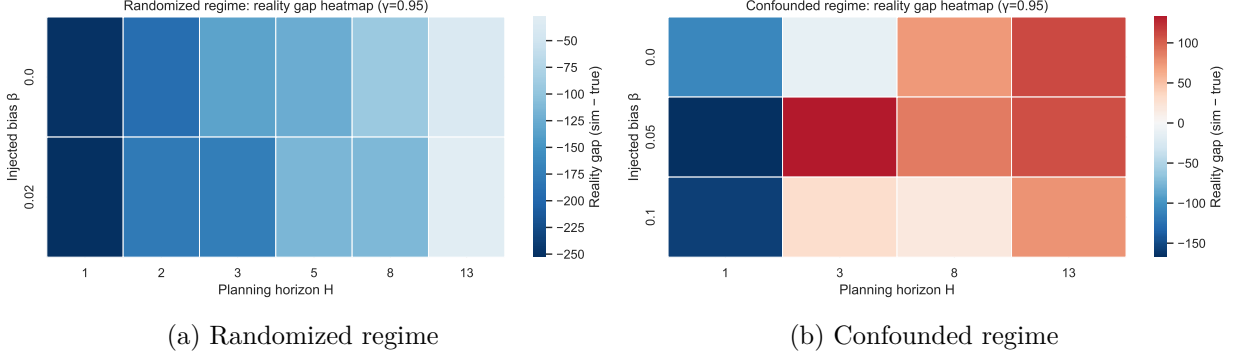


Figure 1: Reality gap heatmaps (simulated minus true return) for MPC as a function of horizon H and injected bias β (figure script fixes γ).

3 Reality gap

We define the reality gap:

$$\Delta(\pi) = J(\pi \mid \hat{P}) - J(\pi \mid P),$$

where $J(\pi \mid \hat{P})$ is the return when rolling out policy π inside the learned model and $J(\pi \mid P)$ is the return when executing π in the ground-truth environment.

4 Results

Table 1 reports mean reality gaps for MPC policies on a matched subset ($H \in \{1, 3, 8, 13\}$, $\gamma \in \{0.95, 0.99\}$). Under randomized training, gaps are largely negative and shrink toward zero with horizon in this subset (conservative simulation). Under confounded training, gaps become positive at longer horizons, indicating horizon-dependent optimism.

5 Practitioner takeaway

World-model planning is most useful for CLV optimization when training data provide credible coverage (e.g., randomized exploration/holdouts). Under confounded logs, long-horizon planning can become systematically optimistic (simulated value exceeds realized value), so simulated ROI should be treated cautiously unless validated online.

References

Table 1: Mean reality gap $\Delta = J(\pi \mid \hat{P}) - J(\pi \mid P)$ for MPC policies (subset). Positive means optimistic simulation (overestimation).

Regime	γ	β	H	Mean reality gap
Randomized (subset shown, $\beta = 0.0$)				
Randomized	0.95	0.00	1	-250.00
Randomized	0.95	0.00	3	-134.49
Randomized	0.95	0.00	8	-93.63
Randomized	0.95	0.00	13	-39.70
Randomized	0.99	0.00	1	-250.00
Randomized	0.99	0.00	3	-148.95
Randomized	0.99	0.00	8	-104.57
Randomized	0.99	0.00	13	10.78
Confounded				
Confounded	0.95	0.00	1	-108.60
Confounded	0.95	0.00	3	-13.98
Confounded	0.95	0.00	8	72.46
Confounded	0.95	0.00	13	110.12
Confounded	0.95	0.05	1	-167.42
Confounded	0.95	0.05	3	132.26
Confounded	0.95	0.05	8	85.60
Confounded	0.95	0.05	13	107.12
Confounded	0.95	0.10	1	-157.80
Confounded	0.95	0.10	3	29.86
Confounded	0.95	0.10	8	17.14
Confounded	0.95	0.10	13	74.84
Confounded	0.99	0.00	1	-108.60
Confounded	0.99	0.00	3	-9.00
Confounded	0.99	0.00	8	5.58
Confounded	0.99	0.00	13	99.34
Confounded	0.99	0.05	1	-167.42
Confounded	0.99	0.05	3	111.54
Confounded	0.99	0.05	8	62.58
Confounded	0.99	0.05	13	104.90
Confounded	0.99	0.10	1	-157.80
Confounded	0.99	0.10	3	-21.02
Confounded	0.99	0.10	8	95.04
Confounded	0.99	0.10	13	147.16